INCH-POUND

MIL-S-47053A(MI)
25 July 1990
SUPERSEDING
MIL-S-46053(MI)
10 MAY 1974

# MILITARY SPECIFICATION

# SLEEVING, TEXTILE, BRAIDED, POLYAMIDE

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1. Scope. This specification covers one type of tubular braided, black polyamide sleeving. The sleeving is suitable for use within a temperature range of minus 55 degrees Celsius (C) to 105 degrees C.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

AMSC N/A FSC 5970

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SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-C-572 | Cords, Yarns and Monofilaments Organic Synthetic Fiber |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-810 | Environmental Test Methods and Engineering Guidelines |
| MIL-STD-1190 | Minimum Guidelines for Level C Preservation, Packing and Marking |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption is obtained.

3. REQUIREMENTS

3.1 First article sample. Unless otherwise specified (see 6.2), a first article sample shall meet all the requirements of this specification.

3.2 Material.

3.2.1 Composition and construction.

3.2.1.1 Composition. The sleeving shall be braided from 210 denier yarn conforming to MIL-C-572, Type P, Form Y.

3.2.1.2 Construction. Sleeving construction shall be as specified below and in Table I.

3.2.1.2.1 Yarn. Yarn used in the sleeving braid shall be 2 ply with an "S" twist (see 6.3).

3.2.1.2.2 Cordage. Cordage used in the sleeving braid shall be 2 ply, 840 denier, with a "Z" twist (see 6.3).



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3.2.1.2.3 Spliced joints. Spliced joints in the braided sleeving shall be limited to fused joints in yarn or cordage only. Spliced joints shall not appear more often than once in 25 yards of sleeving, or not more often that 20 times in 600 yards.

Table I. Sleeving Dimensions and Construction

| Inside diameter, inch, nominal | Number of ends | Number of carriers | Picks per inch | Wall thickness inch |
|---|---|---|---|---|
| 1/8 | 24 | 24 | 20 ± 2 | |
| 3/16 | 32 | 32 | 18 ± 2 | |
| 1/4 | 44 | 44 | 20 ± 2 | 0.030 + 0.015 |
| 5/16 | 44 | 44 | 18 ± 2 | |
| 3/8 | 48 | 48 | 18 ± 2 | |
| 1/2 | 64 | 64 | 16 ± 2 | |
| 5/8 | 128 | 64 | 12 ± 2 | |
| 3/4 | 128 | 64 | 11 ± 2 | 0.040 + 0.015 |
| 7/8 | 192 | 64 | 10 ± 2 | |
| 1 | 216 | 72 | 10 ± 1 | 0.050 + 0.015 |
| 1-1/4 | 216 | 72 | 9 ± 1 | |
| 1-1/2 | 288 | 72 | 8 ± 1 | |
| 1-3/4 | 288 | 72 | 7 ± 1 | 0.060 + 0.015 |
| 2 | 360 | 72 | 6 ± 1 | |



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3.2.2 Sleeving dimensions. Sleeving dimensions shall be as specified in Table I.

3.2.3 Color. Unless otherwise specified, the sleeving color shall be gloss black using a chrome fast dye.

3.2.4 Braid coverage. The coverage of the braid shall not be less than 85 percent.

3.3 Performance characteristics.

3.3.1 Tensile strength (tenacity). The breaking strength of the braid cord shall not be less than 25 pounds. The breaking strength of fused cord splices shall not be less than 70 percent of the cord breaking strength.

3.4 Environmental conditions.

3.4.1 Fungus resistance. The sleeving shall not support the growth of fungus.

3.5 Workmanship. The sleeving shall be free of visible defects, dirt, grime, grease, and abraded or damaged surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.



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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in paragraph 4.7.1.

4.4 First article inspection. Unless otherwise specified in the contractual document (see 6.2), a sample consisting of 20 feet of sleeving, manufactured under the same conditions as those proposed for subsequent production, shall be subjected to first article inspection. Accomplishment of first article inspection shall be as specified herein. Subsequent units will not be considered for acceptance until Government approval of the first article sample has been obtained. Units subjected to first article inspection shall have successfully passed acceptance inspection. Testing of the first article sample, to determine compliance with the characteristics listed in Table II, shall be conducted in accordance with the corresponding test paragraphs.

Table II. First Article Inspection

| Characteristic | Requirement source | Test paragraph |
|---|---|---|
| Tensile strength | 3.3.1 | 4.7.4.5 |
| Fungus resistance | 3.4.1 | 4.7.5.1 |

4.5 Quality conformance inspection.

4.5.1 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.5.2 Sampling inspection. Sampling inspection to determine compliance with the characteristics specified in Table III, shall be conducted in accordance with the corresponding test and inspection paragraphs.

4.6 Inspection equipment. The inspection equipment for conducting examination and tests shall be as specified in the applicable test methods and procedures paragraphs.



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Table III. Acceptance Inspection

| Characteristic | Requirement source | Test and inspection paragraph |
|---|---|---|
| Composition and construction | 3.2.1 | 4.7.4.1 |
| Sleeving dimensions | 3.2.2 | 4.7.4.2 |
| Color | 3.2.3 | 4.7.4.3 |
| Braid coverage | 3.2.4 | 4.7.4.4 |
| Packaging and packing | 5.1 | 4.8 |
| Marking | 5.2 | 4.8 |
| Workmanship | 3.5 | 4.9 |

4.7 Test methods and procedures.

4.7.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements:

a. Temperature, room ambient (16 to 32 degrees C).

b. Altitude, facility ground.

c. Humidity, facility ambient up to 95 percent relative humidity.

4.7.2 Test sequence. Test sequence within each classification of inspection (first article or quality conformation) shall be at the option of the supplier.

4.7.3 Test specimen preparation. Test specimens shall be cut from the sample specified in 4.5.2.

4.7.4 Tests.

4.7.4.1 Composition and construction. The composition shall be certified by the supplier and the construction shall be determined visually to establish conformance to 3.2.1.



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4.7.4.2 Dimensions. The sleeving wall thickness shall be measured with a paper micrometer or equivalent instrument for compliance with 3.2.2.

4.7.4.3 Color. The sleeving shall be visually examined for compliance with color requirements in 3.2.3.

4.7.4.4 Braid coverage. Three specimens of sleeving, each not less than 6 inches in length, shall be carefully cut and fitted over a smooth surfaced mandrel which has hemispherical ends. The mandrel diameter shall correspond in size to the inside diameter of the sleeving within plus and minus 1/64 inch. The sleeving shall be fitted on the mandrel in a manner where it is not stretched (tightly fitted), loosely fitted, compressed, or gathered. The braid coverage shall than be determined by using the formula below and shall conform to 3.2.4. Applicable values to be substituted in the formula shall be carefully obtained from the mandrel mounted sleeving specimen and the braid coverage computed.

Braid coverage formula:

$$K = (2F - F^2)\,100,$$

where K = percent coverage

F = NP x 0.0234/sine a
a = angle of braid with longitudinal axis of sleeving
Tangent a = 2 (D + 0.0468) P/C
N = number of ends per carrier
P = picks per inch of sleeving length
D = inside diameter of sleeving in inches
C = number of carriers

4.7.4.5 Tensile strength (tenacity). Tensile strength of cord, carefully removed from finished sleeving, shall be conducted in accordance with applicable test procedures of MIL-C-572 to establish compliance with 3.3.1.

4.7.5 Environmental test.

4.7.5.1 Fungus resistance. Testing for fungus resistance shall be in accordance with MIL-STD-810, Method 508.3 to establish conformance to 3.4.1.

4.8 Packaging, packing, and marking. The inspector shall ascertain that the packaging, packing, and marking of the material conforms to this specification.

4.9 Workmanship. The sleeving shall be visually examined for conformance to workmanship requirements specified in 3.5.



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5. PACKAGING

5.1 Packaging and packing. The sleeving shall be packaged and packed in accordance with MIL-STD-1190. Unless otherwise specified in the contract or order, the sleeving shall be separated by size and furnished on spools or reels.

5.2 Marking. Internal and external packaging shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name and address.

b. Manufacturer's designation.

c. Lot number or batch number.

d. Date of manufacture.

e. Number of this specification.

f. Sleeving size and quantity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. The sleeving in accordance with this specification is intended for use as protective sheathing over internal wire and cable bundles and harnesses in electronic equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.1.1).

c. Whether a first article sample is required (see 3.1), and if so, pertinent details.

d. Quantity.

e. Requirement for certification of composition (see 4.5.4.1).

f. Whether sleeving color, other than black, is required (see 3.2.3).

g. Part or Identifying Number (PIN).



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6.3 Definitions. Definitions specified in MIL-STD-810 are applicable to corresponding terminology in this specification.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Part or Identifying Number (PIN). The PIN to be used for sleeving acquired to this specification is created as follows:

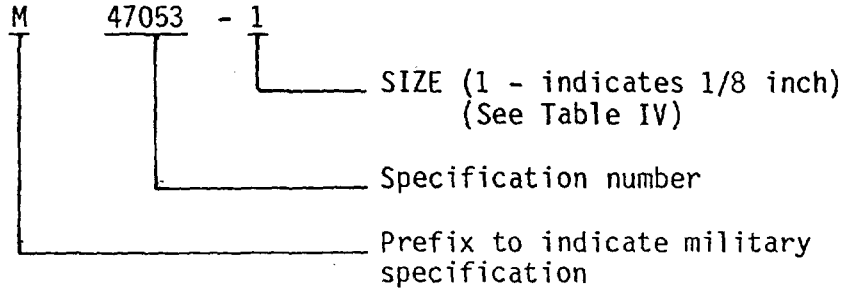



Table IV. Sizes

| Size (inches) | Dash Number |
|---|---|
| 1/8 | 1 |
| 3/16 | 2 |
| 1/4 | 3 |
| 5/16 | 4 |
| 3/8 | 5 |
| 1/2 | 6 |
| 5/8 | 7 |
| 3/4 | 8 |
| 7/8 | 9 |
| 1 | 10 |
| 1 1/4 | 11 |
| 1 1/2 | 12 |
| 1 3/4 | 13 |
| 2 | 14 |



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6.6 Subject term (key word) listing.

Cordage
Sheathing
Yarn

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing Activity:
Army - MI

Project No. 5970-A672



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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-S-47053A(MI) | 900725 |

3. DOCUMENT TITLE
SLEEVING, TEXTILE, BRAIDED, POLYAMIDE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

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| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial | (2) AUTOVON |
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| U.S. Army Missile Command | (205) 876-6980 | 746-6980 |
| c. ADDRESS (Include Zip Code)<br>Commander, U.S. Army Missile Command,<br>ATTN: AMSMI-RD-SE-TD-ST,<br>Redstone Arsenal, AL 35898-5270 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

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